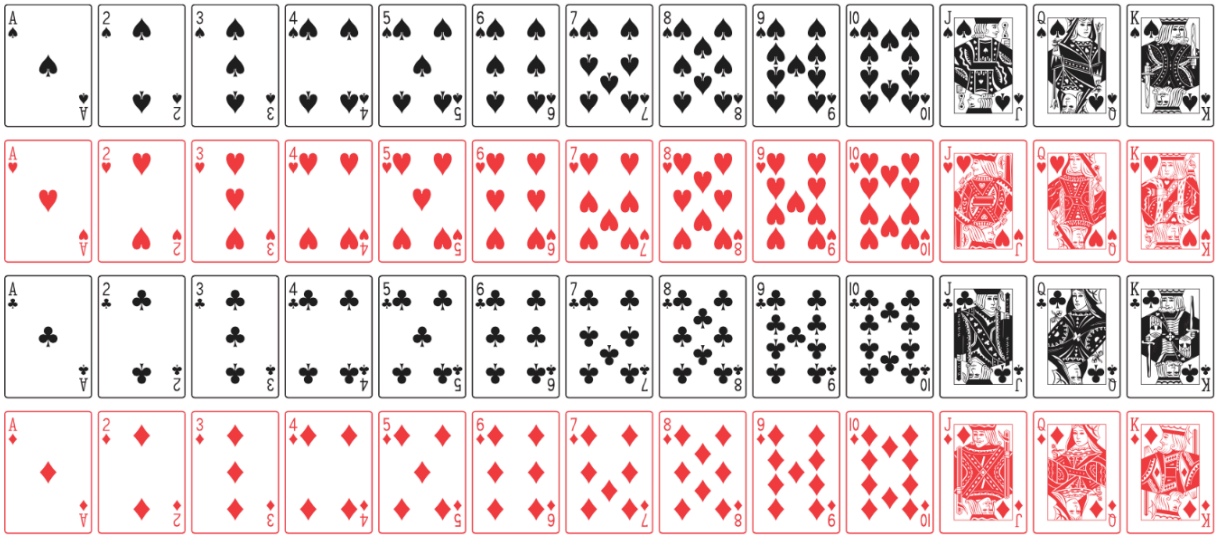
Name: Student ID:

Consider a standard deck of 52 cards, as displayed in the following figure. In the first row (suit: spades ♠) and third row (suit: clubs ♣) all cards are black; in the second row (suit: hearts ♥) and fourth row (suit: diamonds ♦), all cards are red.



Consider the following events when a card is randomly selected.

- $A$: card selected is a heart.
- $B$: card selected is the color black (the spade ♠ and club ♣ cards are black).
- $C$: card selected is a face card (Jack (J), Queen (Q), King (K). Note that Ace (A) does **not** count as a face card.)
- $D$: card selected is a King (K).

1. Find $P(A)$.

2. Find $P(C)$.

3. Find $P\left(D^C\right)$.

4. Find $P(A \cup B)$. Recall that if $A$ and $B$ have outcomes in common, you should only count them **once**.

5. Create a box model to represent drawing a 3 or an 8 in a single draw from the deck of cards described above.

Consider a *seven* sided die (with sides 0, 1, 2, 3, 4, 5 and 6). You roll this die once. Let:

- A be the event that a given roll yields an even number (0 is considered an even number).
- B be the event that a given roll is greater than or equal to three.
- C be the event that the number appears in the phrase "Stat 20".

6. Find $\mathrm{P}(A)$, $\mathrm{P}(B)$ and $\mathrm{P}(C)$. (**Hint**: write out the events *explicitly* and use this to compute their probabilities. That will help with all three questions.)

7. Find $\mathrm{P}(B \cap C)$

8. Find $\mathrm{P}\left(A \cap B^C\right)$.

One number will be drawn at random from each of the two sets below:

$A = \{1, 2, 3\}$; $B = \{1, 2, 3, 4\}$

9. What is the probability the number drawn from $A$ is greater than the one drawn from $B$? (**Hint**: Write out the outcomes, for example $(1, 1)$, $(1, 2)$, ....)

10. What is the probability that the number drawn from $A$ is equal to the one drawn from $B$?

Consider an outcome space $\Omega$ with events $A, B$ that are *not* mutually exclusive. Let $P(A) = 0.5, P(B) = 0.7$ and $P(A \cap B) = .4$. (Is it even *possible* for $A$ and $B$ to be mutually exclusive?)

11. Draw a Venn Diagram containing two circles, one representing $A$ and the other $B$.

12. Shade in the space in the diagram above corresponding to event $A$.

13. Shade in the space in the diagram above corresponding to event $B$ in a way that is different than the way you shaded in the space for event $A$.

14. Based on your shading, calculate the probability $P(A \cup B)$. Explain why you cannot add $P(A)$ and $P(B)$ directly to obtain this answer.